module looks for the :twilio_base_url process dictionary entry. By putting the Bypass URL into the process dictionary (line 10), our API client makes requests to Bypass instead of a real API.

Bypass works by expecting HTTP requests. We use Bypass.expect_once/4 on line 14 to do this. We specify the details of our expected HTTP request and pass a function that will be used to return the HTTP response. In our case, we return a JSON payload (line 21) that's compatible with Twilio's response format.

Based on the response to the Twilio request, our code returns an ok tuple or an error tuple. Our test asserts on both of these cases—line 25 versus line 48. This provides test coverage for our function because it compares both the happy and failure paths.

It may take a while to become comfortable with Bypass. But once you do get comfortable with it, it's incredibly powerful. It's become a critical tool in my personal testing toolbox.

Let's move on to Phoenix endpoint testing next.

Test Phoenix Requests

It's more complex to test endpoints than it is to test pure functions or Ecto queries, but Phoenix provides several helper functions to make it as easy as possible. You'll find more complexity in deciding what to test (complex HTML responses, valid versus invalid parameters, full integration testing versus mock testing, and so on) than you will in writing the actual tests. However, lean toward testing more than less so that you have confidence in your system.

We'll set up the test module and then walk through a few types of controller tests. But first, let's create a little helper to make our tests simpler.

Create Helper Factory

It's common as you write tests to create the same test data over and over again. Usually, setting up data correctly requires multiple lines of code and is susceptible to change as the application develops further. It would be tedious to manually create data again and again, but there's a better way: test factories.

A test factory is a function that initializes data depending on what you need for a particular test. Factories let us set up data without duplicating complex code across our entire application. In our test suite, we're going to use a basic factory function that creates an SMS message.